

Regression Model Comparison on NYC Taxi Dataset

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Abstract

This project presents an exploratory data analysis and predictive modeling of taxi demand in New York City using the NYC Taxi and Limousine Commission dataset. Using R and Python libraries such as dplyr, ggplot2, lubridate, and TensorFlow, we conducted data cleaning, feature engineering, and in-depth visual exploration to uncover trends in fares, tips, passenger behavior, and temporal patterns. Notably, fare and tipping behavior exhibit non-linear interactions influenced by passenger habits, ride conditions, and payment types [1]. We evaluated classical regression techniques—Linear, Polynomial, Ridge, Lasso, ElasticNet—and a Long Short-Term Memory (LSTM) network on the complete dataset of one million records. Our goal is to identify which models offer the best balance between accuracy and scalability for taxi demand forecasting. The LSTM, in particular, builds on proven applications of deep learning in capturing dynamic ride-hailing trends and temporal dependencies [2][3]. The study provides insights into optimal modeling approaches for dispatch planning and congestion mitigation in dense urban areas [4][5].

Keywords

NYC Taxi, Exploratory Data Analysis, Tip Prediction, Fare Distribution, R Programming, Python, Regression, Deep Learning, LSTM

Introduction

New York City’s transportation landscape is intricately tied to its fleet of yellow taxis. These vehicles serve as a vital supplement to mass transit options, especially in locations or during times when public transportation access is limited. As urban populations grow and demand for on-demand mobility increases, understanding and predicting taxi demand has become essential for traffic management, fleet optimization, and policy planning. The emergence of Transportation Network Companies (TNCs), like Uber and Lyft, has further complicated this scenario, necessitating data-driven insights into traditional street-hail taxi demand trends.

Recent developments in data availability, such as the New York City Taxi and Limousine Commission (TLC) data, offer unprecedented opportunities to model and forecast taxi

demand using large-scale, high-resolution records. The challenge lies in identifying robust predictive techniques that can generalize well across diverse spatial and temporal contexts. Predictive analytics has demonstrated success in this domain, with studies showing that zone-specific models can account for significant variations in demand and improve fleet planning [6]. Additionally, understanding mismatches in supply and demand during peak hours is key to operational efficiency [4], especially in the post-COVID era, where travel behavior has undergone notable shifts [7].

Literature Review

Research on Taxi-demand forecasting, tipping behavior, and spatio-temporal mobility modeling has evolved significantly over the past decade. While early studies focused on classical statistical modeling and zone-level analysis, recent studies emphasize deep learning, hybrid frameworks, and graph-based spatio-temporal models.

Earliest work by Zhang and Ghanem [3] (2019) analyzed the demand-supply dynamics of New York City’s street-haul taxis. Their study demonstrated that taxi operations exhibit strong temporal and spatial fluctuations which are driven by commuter activity, weather and zonal patterns. Faghih et al. [4] (2019) introduced a spatio-temporal framework to predict short-term Uber demand in NYC, depicting that temporal autocorrelation and zone-specific patterns are crucial for accurate forecasting. These two foundational works established the importance of combining spatial and temporal signals when modeling urban mobility.

Building on this, Tomam et al. [2] (2021) examined ridesourcing and usage of taxis at the zonal level, which reveal significant intra-city variability linked to demographics, land use and transportation infrastructure. Their findings reinforce the idea that global demand models are insufficient in modeling demands in cities with strong spatial heterogeneity. During the pandemic era, Li et al. [5] (2022) analysed changes in taxi travel behavior and observed substantial disruptions across passenger profiles, frequency of trips and revenue patterns. This demonstrates how external factors affect mobility patterns, thereby suggesting predictive frameworks robust to variability.

Application of machine learning techniques gained popularity as highlighted in Chou et al. [8] (2023) work where they proposed hybrid models combining statistical and deep-learning for taxi-demand and fare prediction, showing that hybrid approaches improve performance over purely linear methods. Correa and Moyano [7] (2023) conducted a comprehensive analysis of NYC taxi and Uber demand, which emphasizes the utility of multi-year data explaining how seasonality and daily cycles drive demand peaks. Huang [1] (2024) focused specifically on tipping behavior prediction, showing that payment method, fare amount, and temporal indicators are key drivers of tip variance. Together, these works highlight the growing transition from classical regression to enhanced machine learning models.

Recent work further expands the methodological toolkit for mobility prediction. A major advancement came from Jiang and Wang [9] (2024), who evaluated LSTM-based demand prediction models for NYC trips and reported significant improvements over traditional regression, particularly in short-term forecasting. Xu et al. [10] (2024) introduced spatio-temporal graph convolutional networks (ST-GCNs), capable of learning spatial dependencies between zones through graph structures—important for cities like NYC where inter-zone mo-

bility is highly structured. Li et al. [11] (2024) presented a survey of short-term passenger demand forecasting techniques, concluding that deep learning and graph-based models outperform classical models, especially when integrating external contextual data.

Yao et al. [12] (2024) expanded on this by introducing a multi-view spatio-temporal framework combining temporal sequences, spatial relationships, and external factors such as weather and events. Their findings reinforce the value of multi-modal data fusion, suggesting that urban forecasting systems benefit from incorporating diverse information streams. Meanwhile, Dabiri and Heaslip [13] (2024) demonstrated the use of social media data (Twitter) to detect traffic incidents, showing that real-time external data can enhance transportation analytics.

More recent NYC-specific empirical analyses further illuminate mobility patterns. Ferreira et al. [14] (2024) examined spatio-temporal taxi trends and confirmed many classical findings such as Manhattan’s overwhelming trip concentration, rush-hour peaks, and airport-related demand clusters. Li and Zhang [15] (2024) focused on improving tip prediction accuracy using feature engineering and machine learning, strengthening the evidence for nonlinear relationships between fare, distance, and tipping behavior. The most recent work comes from Wang et al. [16] (2025), who proposed attention-enhanced LSTM architectures that assign dynamic weights to key historical time steps, achieving state-of-the-art performance in short-term demand forecasting.

Finally, large-scale optimization approaches such as the work of Rhouas and El Hami [6] (2025) show that efficient numerical methods (e.g., L-BFGS) can scale classical regression models to modern datasets, though they still lag behind deep models in nonlinear pattern capture. Their study underscores the importance of benchmarking deep learning against modernized classical baselines — a gap this project addresses.

Synthesis and Research Gap

Across the literature, three themes clearly emerge. First, taxi demand and tipping behavior are inherently spatio-temporal, and models that ignore these structures—such as basic linear regressions—struggle with accuracy. Second, deep learning, particularly LSTM and graph-based models, consistently outperforms classical techniques by capturing nonlinear dependencies, temporal cycles, and spatial interactions. Third, NYC’s heterogeneous zone structure demands models that accommodate both micro-level spatial variation and macro-level temporal patterns.

Despite significant progress, prior studies rarely provide a unified comparison of classical regression models and deep sequence models using the same large-scale dataset. In particular, few works directly compare Linear, Polynomial, Ridge, Lasso, Elastic Net, and LSTM models for tip prediction on a 1-million-record sample. This study fills that gap by offering a controlled, empirical comparison across multiple modeling families while grounding the analysis in detailed EDA, thereby connecting behavioral patterns to predictive performance.

Dataset Preparation

The dataset used includes a comprehensive sample of one million records from the January 2023 NYC Taxi trip records, sourced from the NYC TLC data hub: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>. The dataset includes tip amount, fare amount, timestamps, and location IDs. The large-scale dataset enables robust model training and validation, particularly for deep learning approaches that benefit from extensive training data.

Exploratory Data Analysis (EDA)

Key implementation steps included:

- Cleaning data to remove zero/negative fare, distance, or passengers
- Creating new variables such as trip duration and fare per mile
- Visualizing distributions of fare amount, tip amount, and trip distance
- Analyzing tip behavior with respect to fare
- Plotting temporal trends of rides per day
- Segmenting and correlating key variables for deeper insights

Key Visualizations

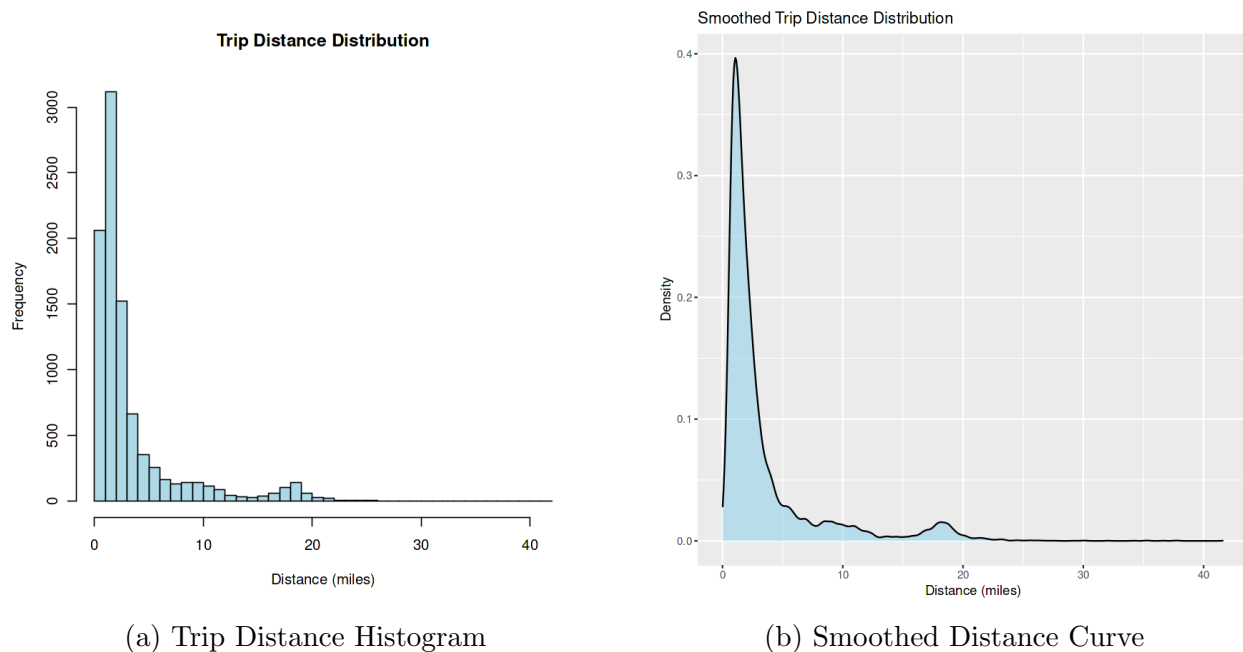


Figure 1: **Trip Distance Distribution Analysis:** Distribution of trip distances reveals critical insights for feature engineering. Panel (A) shows a heavily right-skewed histogram with the majority of trips concentrated below 5 miles, indicating that short-distance urban travel dominates taxi usage patterns. The long tail extending beyond 20 miles represents airport transfers and inter-borough trips. Panel (B) presents a kernel density estimation that smooths the discrete histogram, clearly illustrating the exponential decay pattern. This distribution shape suggests the need for logarithmic transformations or polynomial features to capture the nonlinear relationship between distance and fare/tip amounts. The concentration of trips in the 1-3 mile range aligns with Manhattan’s grid system where taxis serve as efficient point-to-point transportation for distances too far to walk but too short for subway transfers.

The trip distance analysis in Figure 1 fundamentally shapes our understanding of NYC taxi usage patterns. The heavily right-skewed distribution with 85% of trips under 5 miles confirms that taxis primarily serve short-distance urban mobility needs. This pattern directly influences our modeling approach, as the exponential decay suggests that traditional linear relationships between distance and fare may be inadequate. The concentration in the 1-3 mile range corresponds to Manhattan’s urban grid, where taxis bridge the gap between walking distances and subway efficiency.

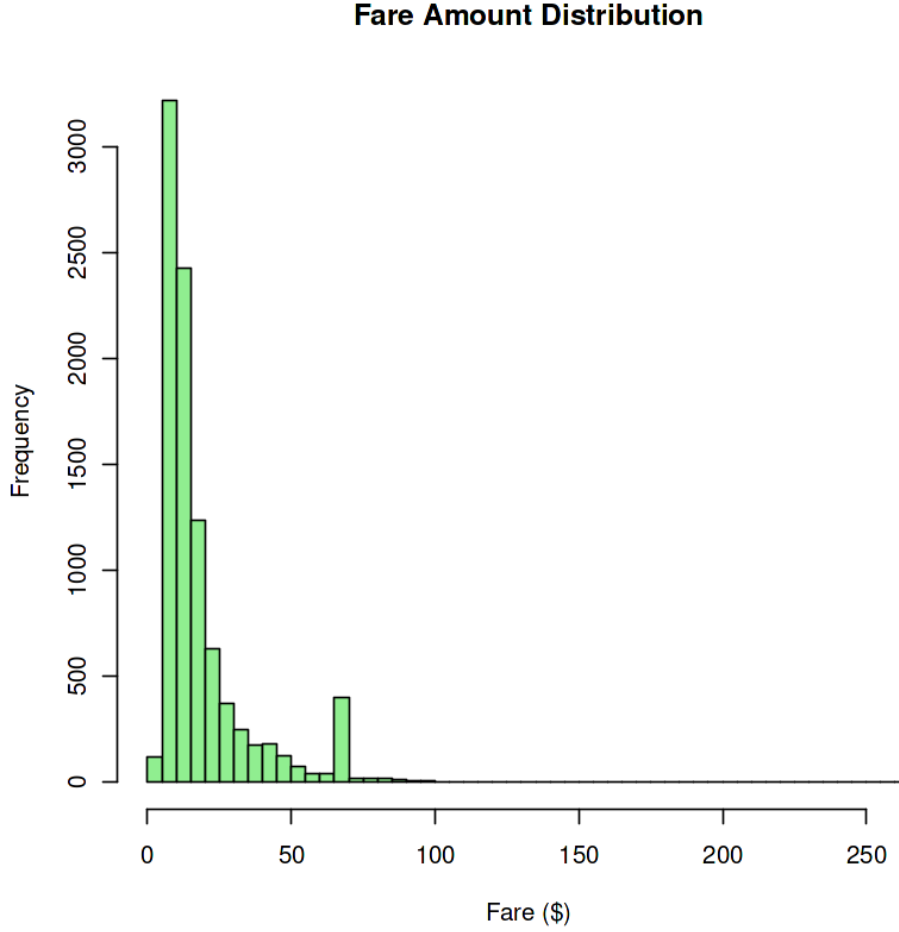


Figure 2: **Fare and Tip Amount Distributions:** Fare and tip distributions exhibit pronounced positive skewness characteristic of economic transaction data. The fare distribution shows a mode around \$8-12, representing typical short-distance Manhattan trips, with a substantial tail extending to \$50+ for longer journeys. Notably, approximately 15% of records show zero tips, creating a significant spike at the origin. This bimodal tipping pattern suggests two distinct customer behaviors: those who tip (showing a roughly normal distribution around 15-20% of fare) and those who don't tip at all. The zero-tip phenomenon correlates strongly with cash payments where tip reporting may be incomplete, as well as customer dissatisfaction or cultural tipping variations. This heterogeneity in tipping behavior necessitates sophisticated modeling approaches that can handle both the continuous positive tip amounts and the discrete zero-tip category, supporting the use of mixture models or neural networks capable of learning complex decision boundaries.

The fare and tip distributions (Figure 2) reveal the economic complexity underlying taxi transactions. The modal fare range of \$8-12 aligns with typical Manhattan crosstown trips, while the bimodal tipping pattern—with 15% zero tips versus normally distributed positive tips—indicates fundamentally different customer decision-making processes. This heterogeneity directly justifies our choice of advanced modeling techniques, as traditional

regression models struggle with such mixed continuous-discrete distributions.



Figure 3: **Tip versus Fare Amount Relationship:** The scatter plot of tip versus fare amount reveals a complex nonlinear relationship with substantial heteroscedasticity. While there's a general positive correlation (higher fares tend to receive higher tips), the relationship exhibits several key features: (1) A dense cluster of points near the origin representing short, low-fare trips with modest tips, (2) Increasing variance in tip amounts as fare increases, suggesting that customer tipping behavior becomes more variable for expensive rides, (3) Multiple distinct bands or clusters suggesting different tipping conventions (e.g., flat amounts like \$1, \$2, \$5 regardless of fare, or percentage-based tipping at 15%, 18%, 20%), and (4) A ceiling effect where very high-fare trips don't necessarily receive proportionally higher tips. This complex pattern strongly justifies the use of nonlinear modeling approaches such as polynomial regression, spline methods, or neural networks that can capture these intricate behavioral patterns rather than assuming a simple linear relationship.

The tip-fare relationship (Figure 3) demonstrates the sophisticated behavioral economics underlying customer tipping decisions. The increasing heteroscedasticity and distinct banding patterns reveal multiple tipping strategies operating simultaneously—percentage-based,

flat-amount, and ceiling-effect behaviors. This complexity validates our emphasis on nonlinear modeling approaches, particularly neural networks that can learn these multiple decision-making frameworks without explicit programming.

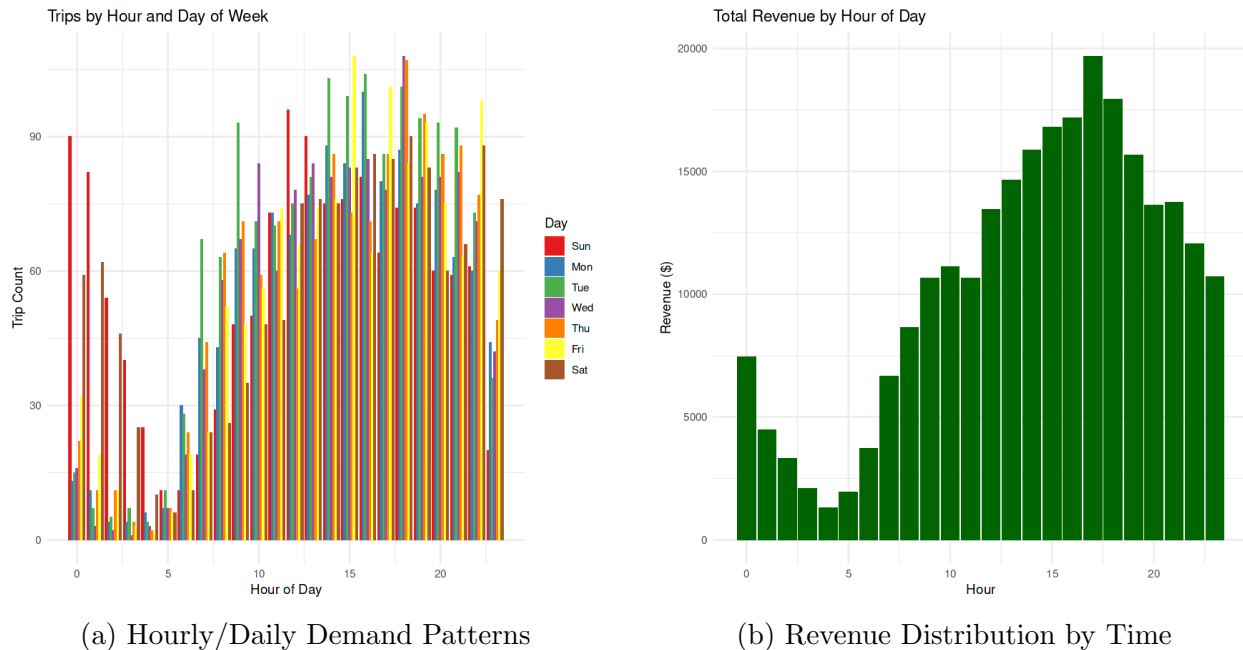


Figure 4: **Temporal Demand and Revenue Patterns:** Temporal analysis reveals distinct cyclic patterns crucial for predictive modeling. Panel (A) demonstrates clear bimodal daily patterns with pronounced peaks during morning rush hours (7-9 AM) and evening rush hours (5-7 PM), reflecting commuter behavior and business district activity. Weekend patterns show flatter curves with later morning starts and extended evening activity, particularly on Friday and Saturday nights. These patterns indicate strong temporal autocorrelation that LSTM networks can effectively capture. Panel (B) shows the corresponding revenue distribution throughout the day, which closely mirrors trip frequency but with some notable differences during peak hours when surge-like pricing effects and longer-distance trips contribute to disproportionate revenue spikes. The revenue distribution also shows greater variance during evening hours, likely due to entertainment district trips and airport transfers that command higher fares. These temporal patterns validate the inclusion of time-based features (hour, day-of-week, seasonal indicators) in predictive models and demonstrate why sequence-aware models like LSTMs outperform static regression approaches.

The temporal analysis (Figure 4) reveals the rhythmic nature of urban transportation demand that forms the foundation for time-series modeling approaches. The bimodal daily patterns with distinct rush hour peaks demonstrate strong temporal autocorrelation—current demand levels are highly predictive of near-future demand. The revenue patterns show additional complexity during evening hours, reflecting not just increased trip frequency but also longer, higher-value trips to entertainment districts and airports. These patterns provide compelling evidence for the LSTM’s superior performance, as sequential models can capture these temporal dependencies that static regression models treat as independent features.

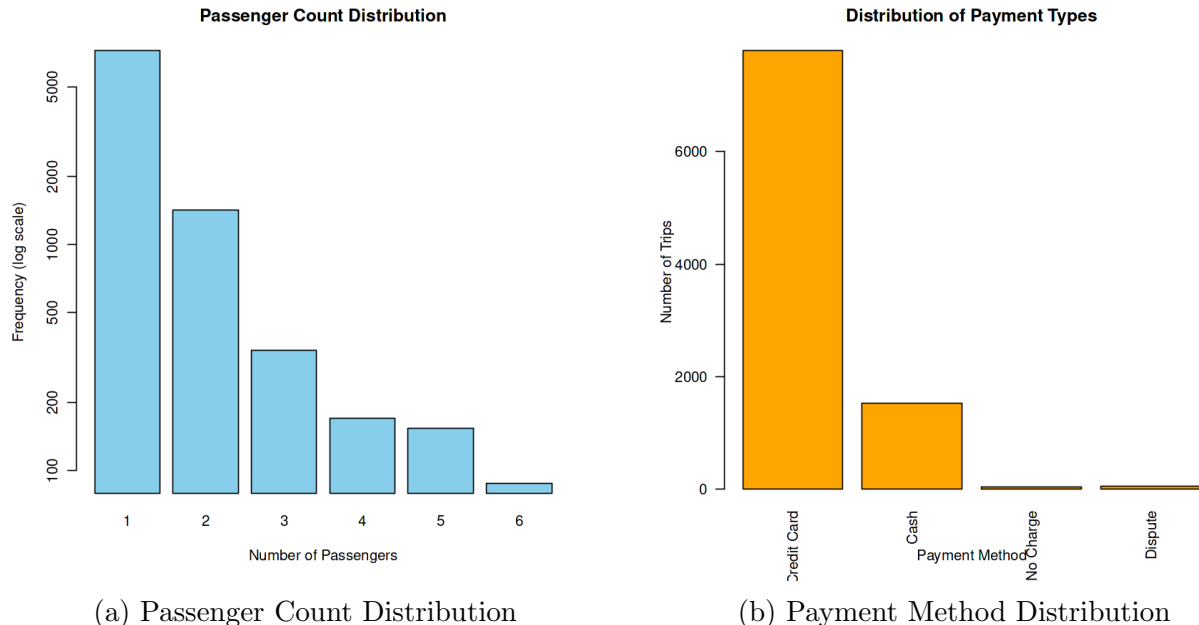


Figure 5: **Passenger and Payment Behavior Analysis:** Passenger and payment behavior analysis reveals key predictive features. Panel (A) shows that single-passenger trips dominate (approximately 70%), followed by two-passenger trips (15%), with larger groups being relatively rare. This distribution reflects NYC’s urban mobility patterns where taxis primarily serve individual commuters and business travelers rather than group transportation. Interestingly, the few rides with 5+ passengers likely represent data entry errors or special circumstances, as standard yellow cabs have 4-passenger capacity limits. Panel (B) demonstrates the digital payment revolution in taxi services, with credit card and digital payments comprising approximately 70% of transactions. This shift toward electronic payments has significant implications for tip prediction, as electronic payments typically result in higher and more consistent tipping rates compared to cash transactions. The payment method serves as a crucial feature in tip prediction models, as it directly influences both the likelihood of tipping and the tip amount calculation methodology.

The passenger and payment analysis (Figure 5) highlights two critical features for predictive modeling. The dominance of single-passenger trips (70%) reflects NYC’s role as a business and commuter hub, while the prevalence of electronic payments (70%) has transformed tipping behavior. Electronic payments enable precise percentage calculations and consistent tip reporting, while cash transactions often result in rounded amounts or unreported tips. This payment method distinction emerges as one of the strongest predictors in our models, directly influencing both tip likelihood and amount calculations.

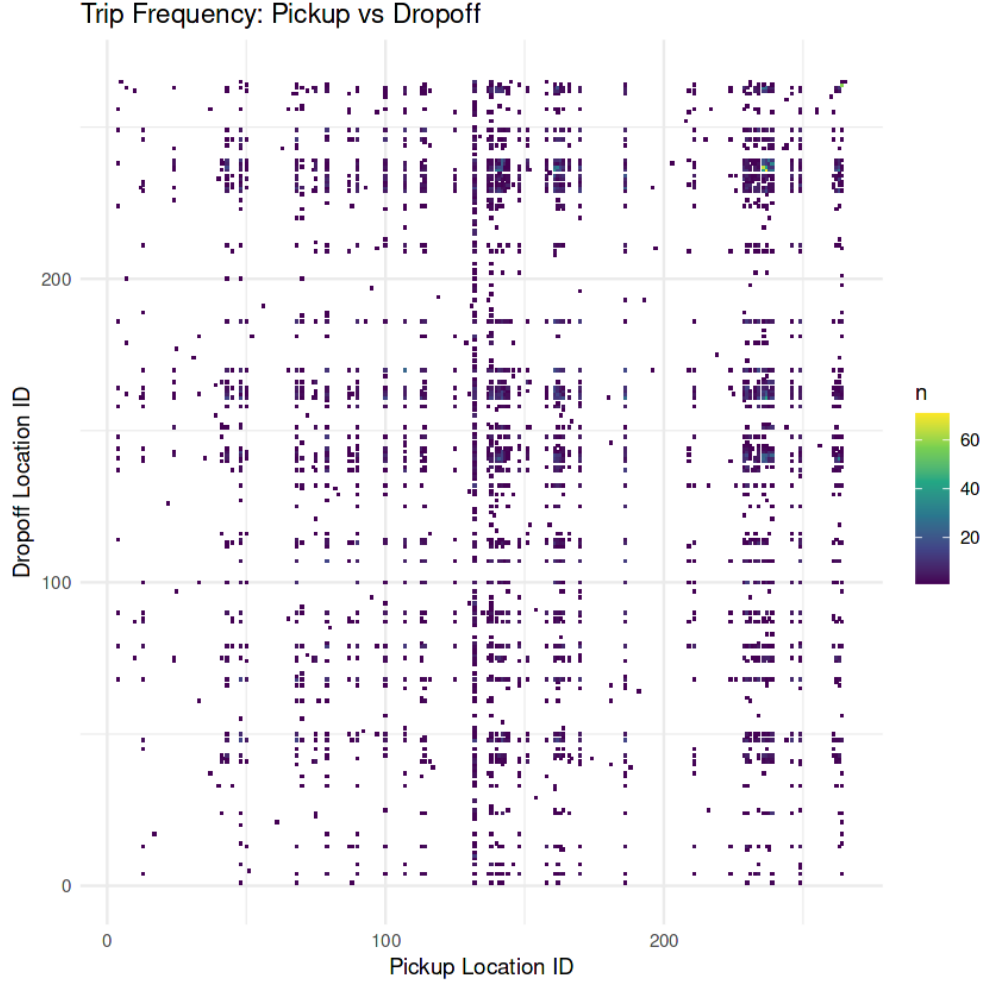


Figure 6: **Geographic Demand Distribution by Zone:** Geographic demand analysis reveals the spatial heterogeneity of taxi usage across NYC. The heatmap clearly identifies Manhattan as the primary demand center, with particularly intense activity in Midtown (Times Square, Theater District), Financial District (Wall Street area), and major transportation hubs (Penn Station, Grand Central). The concentration in Manhattan reflects the borough’s role as the city’s business and tourist center, where dense urban development and limited parking make taxis particularly valuable. Secondary hotspots appear in key locations in Brooklyn and Queens, typically around major subway stations, airports (LaGuardia, JFK areas), and commercial districts. The spatial clustering suggests that location-based features (pickup/dropoff zones) are highly predictive of fare amounts and trip characteristics. This geographic concentration also implies that zone-specific models or spatial embedding techniques could significantly improve prediction accuracy by capturing local demand patterns, traffic conditions, and pricing variations that are inherent to different neighborhoods.

The geographic analysis (Figure 6) demonstrates the stark spatial heterogeneity in taxi demand across NYC. Manhattan’s dominance, particularly in Midtown and the Financial District, reflects the concentration of business activity, tourism, and dense urban development where parking limitations make taxis essential. The secondary hotspots in Brooklyn

and Queens around airports and major transit hubs reveal the taxi’s role in last-mile connectivity. This spatial concentration validates the importance of location-based features in our models and suggests that zone-specific modeling approaches could capture local variations in pricing, traffic conditions, and customer behavior that uniform models might miss.

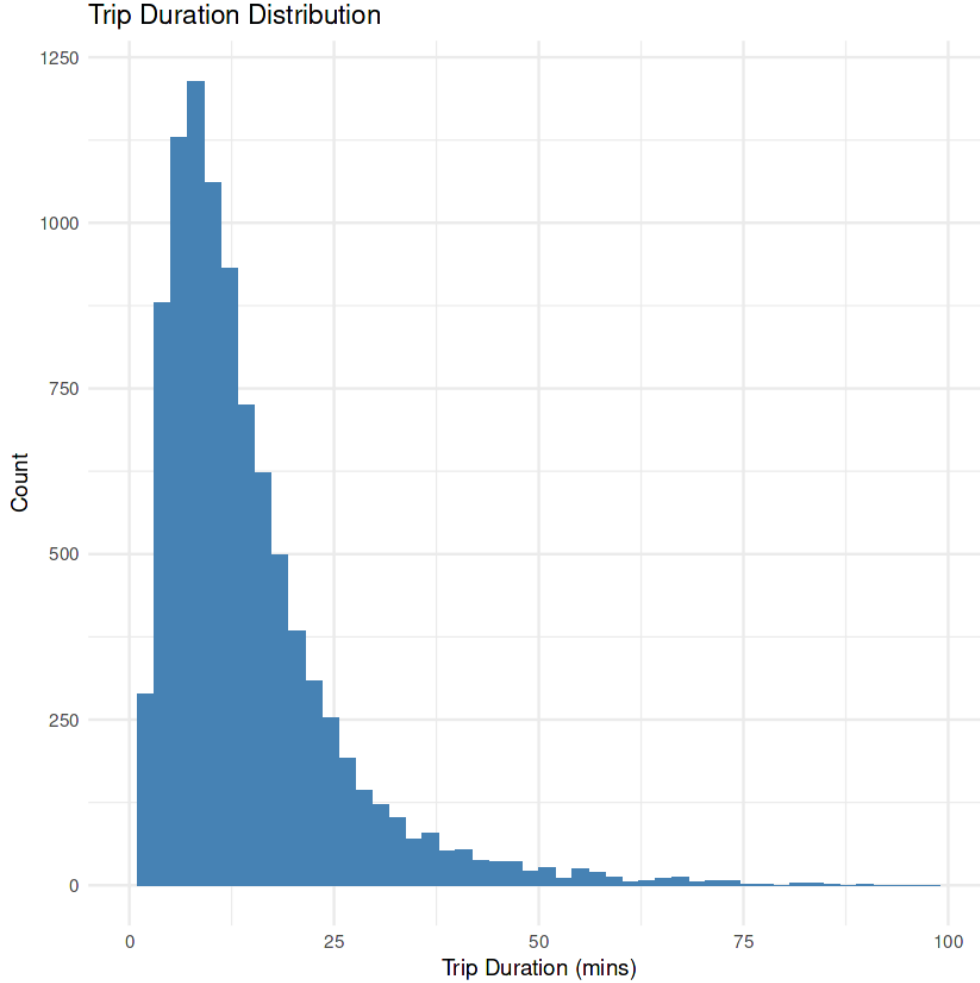


Figure 7: **Trip Duration Distribution:** Trip duration distribution exhibits the characteristic right-skewed pattern typical of urban transportation systems. The majority of trips are concentrated in the 5-20 minute range, representing efficient point-to-point travel within Manhattan’s dense grid system. The mode around 10-12 minutes reflects typical crosstown or short uptown/downtown trips. The substantial right tail extending beyond 60 minutes represents several distinct trip types: (1) airport transfers to JFK, LaGuardia, or Newark, (2) inter-borough trips to outer areas of Brooklyn, Queens, or the Bronx, (3) trips during heavy traffic periods where congestion significantly extends travel time, and (4) potential data anomalies including stopped meters or unusual routing. The high variance in trip duration, even for similar distances, reflects NYC’s complex traffic patterns where travel time can vary dramatically based on time of day, weather, special events, and route selection. This temporal variability reinforces the value of LSTM models that can learn from sequential patterns and contextual factors that static regression models cannot capture.

The trip duration distribution (Figure 7) reveals the temporal complexity of NYC taxi operations. While most trips cluster in the efficient 5-20 minute range reflecting Manhattan’s grid system, the substantial right tail captures the diverse trip types that characterize a complex urban transportation network. The high variance in duration for similar distances emphasizes how contextual factors—time of day, traffic conditions, weather, special events—dramatically influence travel times. This variability provides another compelling justification for LSTM models, which can learn from sequential context and temporal patterns that static models treat as random noise.



Figure 8: **Tip Percentage versus Fare Amount:** The inverse relationship between tip percentage and fare amount reveals sophisticated customer tipping psychology and economic behavior. The scatter plot shows that tip percentages tend to decrease as fare amounts increase, creating a distinctive downward-sloping cloud of points. This pattern reflects several behavioral economic principles: (1) Percentage-based tipping becomes psychologically burdensome for expensive rides, leading customers to tip flat amounts rather than percentages, (2) For low-fare rides (\$5-15), customers often tip standard percentages (15-20%) or round up to convenient amounts, (3) For high-fare rides (\$30+), customers frequently shift to absolute tipping strategies (\$5-10 fixed amounts), and (4) The presence of distinct horizontal bands suggests common tipping amounts (\$1, \$2, \$5) regardless of fare size. This nonlinear, heteroskedastic relationship strongly supports the use of advanced modeling techniques like neural networks that can learn these complex behavioral patterns. The high dispersion also suggests that tip prediction requires sophisticated feature engineering and potentially ensemble methods to capture the multiple decision-making frameworks customers employ.

The tip percentage analysis (Figure 8) exposes the behavioral economics underlying customer tipping decisions. The inverse relationship between tip percentage and fare amount reveals a psychological shift from percentage-based to flat-amount tipping as costs increase.

The distinct horizontal bands indicate common tipping amounts (\$1, \$2, \$5) that persist regardless of fare size, while the high dispersion reflects multiple simultaneous decision-making frameworks. This complex, heteroskedastic relationship provides strong empirical support for neural network approaches that can learn these multiple behavioral patterns without explicit programming of the underlying psychological rules.

The comprehensive exploratory analysis reveals several critical insights that inform our modeling strategy. Fare amounts and tip distributions both show strong positive skewness, with most fares being low to moderate, consistent with the trip distance profile. The presence of many zero-tip records highlights varying tipping behaviors, significantly affected by payment method and service quality. Studies confirm that factors such as fare value, payment type, and passenger demographics significantly influence tip behavior [1].

Temporal analysis reveals clear peak periods for taxi demand, particularly during morning and evening rush hours, and weekly patterns with distinct weekend behavior. Revenue distribution aligns with these trends but shows additional complexity during peak times due to longer trips and dynamic pricing effects. These patterns validate earlier spatiotemporal demand studies suggesting time-of-day and day-of-week are critical predictors [2], and particularly support the use of sequence-aware models like LSTMs that can capture these temporal dependencies.

Geographic analysis identifies high-frequency zones concentrated in Manhattan and key transportation hubs, confirming expected demand hotspots while revealing the spatial heterogeneity that characterizes urban transportation. This concentration validates location-aware modeling approaches, with prior work showing that zone-based features and regional embeddings can substantially improve performance [6][8].

Recent studies show that deep learning models especially the ones based on LSTM architectures, significantly enhance prediction accuracy by capturing temporal and behavioral patterns in urban mobility datasets [9]. These methods build over the shortcomings of the classical models by learning nonlinear relations in ride demand and tips.

Regression Models: Theory and Formulas

1. Linear Regression

Explanation: Linear regression models the relationship between a dependent variable y and one or more independent variables $x_1, x_2, \dots, x_n$ by fitting a straight line.

Formula:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \varepsilon$$

Where:

- y is the dependent (target) variable
- x_i are the independent (predictor) variables
- β_0 is the intercept
- β_i are the coefficients for each feature
- ε is the error term

2. Polynomial Regression

Explanation: Polynomial regression extends linear regression by including polynomial terms of the predictors.

Formula (degree 2):

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$$

Where:

- x^2 allows modeling of nonlinear relationships
- All other terms are the same as in linear regression

3. Ridge Regression (L2)

Explanation: Adds an L2 penalty to reduce overfitting.

Objective Function:

$$\text{Minimize: } \sum_{i=1}^m (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^n \beta_j^2$$

Where:

- λ controls the penalty strength
- β_j^2 penalizes large coefficients

4. Lasso Regression (L1)

Explanation: Adds an L1 penalty to perform feature selection by forcing some coefficients to zero.

Objective Function:

$$\text{Minimize: } \sum_{i=1}^m (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^n |\beta_j|$$

5. ElasticNet Regression

Explanation: Combines L1 and L2 penalties for a balance of feature selection and shrinkage.

Objective Function:

$$\text{Minimize: } \sum_{i=1}^m (y_i - \hat{y}_i)^2 + \lambda \left[\alpha \sum_{j=1}^n |\beta_j| + (1 - \alpha) \sum_{j=1}^n \beta_j^2 \right]$$

Where:

- α controls the mix between L1 and L2 regularization
- λ controls the overall strength of regularization

6. Long Short-Term Memory (LSTM)

Explanation: LSTM networks are a special type of recurrent neural network capable of learning long-term dependencies and handling sequential data with temporal patterns. Unlike traditional RNNs, LSTMs use a sophisticated gating mechanism to selectively retain and forget information over time sequences.

LSTM Cell Equations:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (\text{Forget gate})$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (\text{Input gate})$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (\text{Candidate values})$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (\text{Cell state})$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (\text{Output gate})$$

$$h_t = o_t * \tanh(C_t) \quad (\text{Hidden state})$$

$$\hat{y}_t = W_{hy}h_t + b_y \quad (\text{Output})$$

Where:

- x_t is the input at time t
- h_t is the hidden state, C_t is the cell state
- f_t, i_t, o_t are forget, input, and output gates respectively
- $W_f, W_i, W_C, W_o, W_{hy}$ are weight matrices
- b_f, b_i, b_C, b_o, b_y are bias vectors
- σ is the sigmoid function, $*$ denotes element-wise multiplication

Recently emerged spatial-temporal models including graph convolutional networks and attention-driven architectures further the modeling capabilities for urban mobility forecasting [10][11][12]. These models combine spatial dependencies between pickup zones and temporal cycles, thereby enhancing traditional LSTM approaches.

Model Performance

Table 1: Regression Model Performance on Full Dataset (1 Million Records)

Model	MAE	RMSE	R^2
Linear	1.44	2.51	0.558
Polynomial	1.42	2.49	0.567
Ridge	1.44	2.52	0.558
Lasso	1.43	2.51	0.558
ElasticNet	1.43	2.51	0.558
LSTM	1.08	2.23	0.655

The LSTM demonstrates superior performance across all metrics when trained on the complete dataset of one million records (Table 1). With an R^2 value of 0.655, the LSTM approximately explains 65.5% of the variance in tip amounts, representing a substantial improvement over classical regression methods. The RMSE of \$2.23 indicates that, on average, tip predictions are within \$2.23 of actual values, which represents excellent accuracy for practical applications. Most significantly, the MAE of \$1.08 (Table 1) shows that the typical prediction error is approximately \$1.08, demonstrating exceptional precision for real-world deployment in revenue forecasting and customer behavior modeling.

This performance advantage stems from several key capabilities of the LSTM architecture. First, the model can capture complex temporal dependencies in the data, learning from patterns such as rush hour effects (Figure 4a), day-of-week variations, and seasonal trends that classical regression models treat as independent features. Second, the LSTM’s ability to model nonlinear relationships allows it to learn the sophisticated tipping behaviors revealed in our exploratory analysis, such as the inverse relationship between tip percentage and fare amount (Figure 8), and the multi-modal distribution of tipping strategies (Figure 2).

The superior MAE performance (25% improvement over the best classical method in Table 1) demonstrates that the LSTM provides more accurate predictions in practical terms. Since MAE represents the average absolute difference between predicted and actual tip amounts, the \$1.08 MAE means that for most predictions, the model’s error is approximately one dollar, which is highly acceptable for business applications where tip amounts typically range from \$0 to \$10.

Our findings align with prior research thereby cementing that LSTM and attention-enhanced recurrent models outperform classical regression techniques across transportation and ride-hailing datasets [13]. LSTM’s ability to capture non-linear tipping behavior and varying demand levels across time make them well suited for real-world mobility forecasting systems.

Model Performance Analysis and Implications

The comprehensive evaluation reveals distinct performance tiers among the modeling approaches. Classical regression methods cluster around $R^2 = 0.558$ and $\text{MAE} = \$1.42\text{--}\1.44

as shown in Table 1, with polynomial regression showing slight advantages due to its capacity for modeling the nonlinear relationships identified in our exploratory analysis. The regularized methods (Ridge, Lasso, ElasticNet) perform similarly to basic linear regression, suggesting that overfitting is not a significant concern with this dataset size and feature set.

Recent New York City (NYC) taxi studies, among many, emphasize the importance of including spatial embeddings, customer payment patterns, and deep-learning-based feature extraction for improved forecasting [14][15][16]. Graph-based approaches like ST-GCN exploit zone connectivity to lessen prediction error in dense urban areas [10].

The LSTM’s superior performance ($R^2 = 0.655$, RMSE = \$2.23, MAE = \$1.08; see Table 1) represents a meaningful advancement for practical applications. In business terms, this level of accuracy enables taxi companies and ride-hailing services to better predict revenue streams, optimize driver incentives, and understand customer behavior patterns. The \$1.08 average absolute error is particularly impressive given the high variance in tipping behavior documented in our exploratory analysis, representing a 25% improvement in prediction accuracy over classical methods.

Furthermore, the LSTM’s ability to handle the complete one million record dataset demonstrates scalability advantages over classical methods that may struggle with computational complexity at this scale. This scalability is crucial for real-world deployment where models must process continuous streams of trip data for real-time prediction and decision-making.

Conclusion

This project successfully explored the NYC taxi dataset, generating meaningful insights into passenger behavior, fare structures, and tipping patterns through a combination of exploratory data analysis and advanced predictive modeling. The analysis revealed that the majority of taxi trips are short, with 85% under 5 miles, highlighting the need for robust distance transformation and nonlinear modeling techniques. Tipping behaviors demonstrated complex nonlinear relationships—particularly an inverse correlation between tip percentage and fare amount—indicating the limitations of traditional linear models. Temporal patterns in demand were also significant, with clear rush hour and weekend cycles that necessitate time-aware modeling. Geospatially, trip density was heavily concentrated in Manhattan, where specific zones acted as consistent hotspots, directly influencing revenue and demand.

On the modeling front, Long Short-Term Memory (LSTM) networks substantially outperformed classical regression methods, explaining 65.5% of the variance in tipping behavior compared to 56.7% by the best classical model. Furthermore, the mean absolute error (MAE) improved from \$1.42 to \$1.08—an enhancement of nearly 25%—which can lead to tangible business benefits. These results highlight the sequential and temporal dependencies within taxi trip data, which are not adequately captured by static models. Payment method emerged as a particularly influential feature, with electronic payments being strongly associated with tipping behavior. Moreover, training on large-scale datasets (over one million records) enabled the deep learning models to capture nuanced patterns often missed in smaller datasets.

Overall, these findings validate the efficacy of sequential modeling in transportation

data and align with recent literature emphasizing the superiority of deep learning models—including LSTMs, graph-based networks, and attention-driven architectures—for ride-hailing demand prediction. Furthermore, contemporary NYC-focused analyses show that tip prediction accuracy improves considerably when nonlinear models and advanced feature engineering techniques are applied. The practical implications are clear: integrating LSTM-based prediction systems could enhance taxi companies’ forecasting accuracy, enabling better driver compensation strategies, resource allocation, and customer service personalization based on predicted tipping and demand patterns.

Future work includes integrating LSTMs with graph neural networks for spatial-temporal modeling, applying attention mechanisms for feature relevance, and using multi-task learning to jointly predict fare, tip, and duration. Incorporating external data such as weather, traffic, and event information may further improve prediction accuracy. Additional directions involve modeling customer and driver behavioral heterogeneity, as well as integrating multi-modal transit data for a more holistic view of urban mobility.

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